University Of Engineering and Technology, Lahore
Computer Engineering Department

Lab Evaluation:

Course Name: Data Structures & Algorithm	**Course Code:** CMPE-212L
Assignment Type: Lab	**Dated:** 19th March 2024
Semester: 4th	**Session:** 2022
Lab/Project/Assignment #: 8	**CLOs to be covered:** CLO 1
Lab Title: Sorting Algorithm	**Teacher Name:** Engr. Afeef Obaid

CLO 1	Evaluate known data structures and algorithm for problem solving.					
Levels (Marks)	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
(10)						
					Total	**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	**9-10**	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	**7-8**	L2	Submitted the lab tasks but have good understanding
Good	**5-6**	L3	Submitted the lab tasks but have weak understanding.
Basic	**3-4**	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	**1-2**	L5	Submitted only one lab task.
Not Acceptable	**0**	L6	Did not attempt

LAB No. 8

Lab Goals/Objectives:

- Identify the running time of various sorting algorithms on preset data

Equipment Required: Computer and Dev C++ Compiler

Theory/Relevant Material: Sorting Algorithms

Lab Tasks:

- Code the following sorting algorithms and find their hidden constants. For sorting you are given data files containing average and worst-case data values, respectively. Size of data to be sorted is, 2^{10}.

  Sorting algorithms

  ● Insertion Sort

  ● Merge Sort

  ● Bubble sort

  ● Quick Sort

  Plot your results to observe if the trend of running time matches that of calculated Asymptotical run time.